Interdisciplinarity Formula Cheatsheet

Diversity and disparity

- **Rao–Stirling diversity (Δ; §§3.1, 4.1)**

$$\Delta = \sum_{i \neq j} d_{ij}\, p_i\, p_j = 1 - \sum_{i,j} s_{ij}\, p_i\, p_j$$

  with $d_{ij} = 1 - s_{ij}$ from the category similarity matrix.

- **Hill-type true diversity (${}^{q}D^{S}$; §3.1)**

$${}^{q}D^{S} = \left(\sum_{i=1}^{N} p_i \left(\sum_{j=1}^{N} s_{ij}\, p_j \right)^{q-1} \right)^{1/(1-q)}$$

  similarity-sensitive Hill number with sensitivity parameter q.

- **Quadratic true diversity (${}^{2}D^{S}$; §3.1)**

$${}^{2}D^{S} = \frac{1}{\sum_{i,j} s_{ij}\, p_i\, p_j}$$

  effective-number interpretation; related to Rao–Stirling via ${}^{2}D^{S} = 1/(1-\Delta)$.

- **Herfindahl concentration (toy example, §4.2)**

$$H_A = 0.300^2 + 0.075^2 + 0.175^2 + 0.350^2 + 0.100^2 = 0.258750$$

  sample computation for Researcher A's portfolio.

Coherence and network structure

- **Bibliographic coupling similarity (s_{ab}; §3.2)**

$$s_{ab} = \frac{\mathbf{r}_a \cdot \mathbf{r}_b}{\|\mathbf{r}_a\| \, \|\mathbf{r}_b\|}$$

  Salton cosine between reference vectors $\mathbf{r}_a$ and $\mathbf{r}_b$.

- **Mean linkage strength (S; §§3.2, 4.1, robustness)**

$$S = \frac{2}{N(N-1)} \sum_{a<b} s_{ab}$$

$$S = \frac{1}{\binom{n}{2}} \sum_{k<l} \cos(\mathbf{r}_k, \mathbf{r}_l) = \frac{1}{\binom{n}{2}} \sum_{k<l} \frac{\mathbf{r}_k \cdot \mathbf{r}_l}{\|\mathbf{r}_k\| \, \|\mathbf{r}_l\|}$$

  average bibliographic-coupling strength across a publication set.

- **Betweenness centrality (g_i; §3.2)**

$$g_i = \sum_{\substack{j,k \\ j \neq k \neq i}} \frac{g_{ijk}}{g_{jk}}$$

  Freeman betweenness for positional bridging in journal networks.

- **Citing-distribution coherence (C; §3.3)**

$$C = \sum_{i \neq j} p_{ij} \, d_{ij}$$

  average cognitive distance among co-occurring citation categories.

- **DIV indicator (cited direction; §3.3)**

$$\mathrm{DIV}_c = \frac{n_c}{N} \, (1 - G_c) \, \frac{\sum_{i \neq j} d_{ij}}{n_c(n_c - 1)}$$

  variety–balance–disparity decomposition for citations received.

- **Gini coefficient for balance (G_c; §3.3)**

$$G = \frac{\displaystyle\sum_{i=1}^{n}(2i - n - 1)\, x_i}{\displaystyle n\sum_{i=1}^{n} x_i}$$

  applied to citation-share vector x sorted in non-decreasing order.

- **Cross-field effect proxy (E; §4.1)**

$$E = \frac{\text{citations from articles whose primary category} \neq k^*}{\text{total citations received}}$$

  share of citations arriving from outside each paper's primary category.

Novelty measures

- **Generic divergence framing (§3.4)**

$$\nabla = f\big(|p(x) - p(E)|\big).$$

- **Probabilistic Jaccard divergence (∇_{PJ}; §3.4)**

$$\nabla_{\mathrm{PJ}}(x) = 1 - \frac{\sum_i \min\big(p_i(x),\, p_i(E)\big)}{\sum_i \max\big(p_i(x),\, p_i(E)\big)}.$$

- **Hellinger distance (∇_{Hel}; §3.4)**

$$\nabla_{\mathrm{Hel}}(x) = \frac{1}{\sqrt{2}}\sqrt{\sum_i \Big(\sqrt{p_i(x)} - \sqrt{p_i(E)}\Big)^2}.$$

- **Uzzi atypicality z-score (∇_{Uzzi}; §3.4)**

$$\nabla_{\mathrm{Uzzi}}(p_{i,j},\, x) = \frac{p_{i,j}(x) - e_{i,j}(x)}{\sigma\big[e_{i,j}(x)\big]}, \quad p_{i,j}(x) > 0.$$

- **Timed novelty (§3.4)**

$$\mathrm{Novelty}(x) = \sum_{i,j} t_0(i,j) \cdot \big[1 - z(i,j)\big],$$

weighting first-seen combinations by disparity.

Similarity-perturbation robustness (diversity; §4.3)

- **Uniform perturbation effect on Δ**

$$\Delta_{\text{new}} = \Delta_{\text{old}} - \varepsilon\,(1 - H)$$

with $H = \sum_i p_i^2$.

- **Gap evolution under perturbation**

$$\Delta_{A,\text{new}} - \Delta_{B,\text{new}} = (\Delta_A - \Delta_B) + \varepsilon\,(H_A - H_B)$$

vanishes at

$$\varepsilon^* = -\frac{\Delta_A - \Delta_B}{H_A - H_B}, \qquad \varepsilon^* = \frac{0.020625}{0.058750} = 0.35106.$$

- **A/B vs. specialist separation**

$$\overline{\Delta}_{AB} - \Delta_C = \overline{\Delta}_{AB,0} - \Delta_{C,0} - \varepsilon\,\big(\overline{1 - H}_{AB} - (1 - H_C)\big).$$